

This document is a **transcript** of the lecture, with extra summary and vocabulary sections for your convenience. Due to time constraints, the notes may sometimes only contain limited illustrations, proofs, and examples; for a more thorough discussion of the course content, **consult the textbook**.

Summary

Quick summary of today's notes. Lecture starts on next page.

- If A and B are $n \times n$ matrices with $AB = I_n$ then $BA = I_n$ and $A^{-1} = B$.
- A **subspace** H of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ containing the zero vector that is closed under linear combinations. This means that $0 \in H$ and if $u, v \in H$ and $c \in \mathbb{R}$ then $u + v \in H$ and $cv \in H$.
- The **zero subspace** of $\mathbb{R}^n$ is the set $\{0\}$ with just the zero vector $0 \in \mathbb{R}^n$. Let A be an $m \times n$ matrix. The **column space** of A is the span of the columns of A . Denoted $\text{Col } A$. This is a subspace of $\mathbb{R}^m$.

$$\text{Col} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbb{R}\text{-span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} a \\ b \\ a \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\} \subseteq \mathbb{R}^4$$

The **null space** of A is the set of vectors $\text{Nul } A = \{v \in \mathbb{R}^n : Av = 0\}$. This is a subspace of $\mathbb{R}^n$.

$$\text{Nul} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x = y + 2z = 0 \right\} = \left\{ \begin{bmatrix} 0 \\ -2z \\ z \end{bmatrix} : z \in \mathbb{R} \right\} \subseteq \mathbb{R}^3.$$

- A **basis** for a subspace $H \subseteq \mathbb{R}^n$ is a linearly independent spanning set.
- The **standard basis** of $\mathbb{R}^n$ is $e_1, e_2, \dots, e_n$ where $e_i \in \mathbb{R}^n$ is the vector with 1 in row i and 0 in all other rows. Any subspace of $\mathbb{R}^n$ has a basis with at most n vectors.
- The pivot columns of an $m \times n$ matrix A form a basis for $\text{Col } A$.
- Both A and $\text{RREF}(A)$ have the same null space. Usually $\text{Col } A \neq \text{Col } \text{RREF}(A)$.

To find a basis for $\text{Nul } A$, determine the indices $i_1, i_2, \dots, i_p$ of the non-pivot columns of A .

Then there are unique vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ such that any

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad \text{with} \quad \text{RREF}(A)x = 0$$

can be written as $x = x_{i_1}v_1 + x_{i_2}v_2 + \dots + x_{i_p}v_p$. The vectors $v_1, v_2, \dots, v_p$ are a basis for $\text{Nul } A$.

For example, if $\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & 4 & -1 \\ 0 & 0 & 1 & 0 & 2 \end{bmatrix}$ then any $x \in \mathbb{R}^5$ with $\text{RREF}(A)x = 0$ has

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 4x_4 + x_5 \\ x_2 \\ -2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \\ 1 \end{bmatrix}.$$

The three vectors on the right are a basis for $\text{Nul } A = \text{Nul } \text{RREF}(A)$.

1 Last time: inverses

The following all mean the same thing for a function $f : X \rightarrow Y$:

1. f is *invertible*.
2. f is one-to-one and onto.
3. For each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.
4. There is a unique function $f^{-1} : Y \rightarrow X$, called the *inverse* of f , such that

$$f^{-1}(f(a)) = a \quad \text{and} \quad f(f^{-1}(b)) = b \quad \text{for all } a \in X \text{ and } b \in Y.$$

The standard matrix of a linear function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the $m \times n$ matrix A with $T(v) = Av$ for all $v \in \mathbb{R}^n$.

Proposition. If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear and invertible then $m = n$ and T^{-1} is linear and invertible.

The following all mean the same thing for an $n \times n$ matrix A :

1. A is *invertible*.
2. A is the standard matrix of an invertible linear function $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$.
3. There is a unique $n \times n$ matrix A^{-1} , called the *inverse* of A , such that

$$A^{-1}A = AA^{-1} = I_n \quad \text{where we define } I_n = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}.$$

4. For each $b \in \mathbb{R}^n$ the equation $Ax = b$ has a unique solution.
5. $\text{RREF}(A) = I_n$
6. The columns of A are linearly independent and their span is $\mathbb{R}^n$.

Proposition. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 matrix.

- (1) If $ad - bc = 0$ then A is not invertible.
- (2) If $ad - bc \neq 0$ then $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Proposition. Let A and B be $n \times n$ matrices.

1. If A is invertible then $(A^{-1})^{-1} = A$.
2. If A and B are both invertible then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is invertible then A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Process to compute A^{-1}

Let A be an $n \times n$ matrix. Consider the $n \times 2n$ matrix $\begin{bmatrix} A & I_n \end{bmatrix}$.

If A is invertible then $\text{RREF}(\begin{bmatrix} A & I_n \end{bmatrix}) = \begin{bmatrix} I_n & A^{-1} \end{bmatrix}$.

So to compute A^{-1} , row reduce $\begin{bmatrix} A & I_n \end{bmatrix}$ to reduced echelon form, and then take the last n columns.

→ If A is not invertible then we will still have a row reduced form but not the identity matrix.

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 4 & 0 & 5 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 0 \\ 4 & 0 & 5 & 0 & 0 & 1 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & -3 & -4 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 & -\frac{4}{3} & 0 & -\frac{1}{3} \end{array} \right] \rightarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{3} & 0 & \frac{2}{3} \\ 0 & 1 & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 & \frac{4}{3} & 0 & -\frac{1}{3} \end{array} \right]$$

$\underbrace{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}}_{I_3} \quad \underbrace{\begin{matrix} -\frac{5}{3} & 0 & \frac{2}{3} \\ 0 & \frac{1}{3} & 0 \\ \frac{4}{3} & 0 & -\frac{1}{3} \end{matrix}}_{A^{-1}}$

$$A^{-1} = -\frac{1}{3} \begin{bmatrix} 5 & 0 & -2 \\ 0 & -1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

Before $A \Leftrightarrow B+C$ Now: $A \Leftrightarrow B \Leftrightarrow C$

2 Stronger characterization of invertible matrices

Remember that a matrix can only be invertible if it has the same number of rows and columns.

Theorem. When A is a square $n \times n$ matrix, the following are equivalent:

- (a) A is invertible. ↪ only applies to square matrix.
 (b) The columns of A are linearly independent.
 (c) The span of the columns of A is $\mathbb{R}^n$

Proof. We already know that (a) implies both (b) and (c).

Assume just (b) holds. Then A has a pivot position in every column, so $\text{RREF}(A) = I_n$ since A has the same number of rows and columns. But this implies that A is invertible.

Similarly, if (c) holds then A has a pivot position in every row, so $\text{RREF}(A) = I_n$ and A is invertible. $\square$

Corollary. Suppose A and B are both $n \times n$ matrices. If $AB = I_n$ then $BA = I_n$.

This means that if we want to show that $B = A^{-1}$ then it is enough to just check that $AB = I_n$.

Proof. Assume $AB = I_n$. Then the columns of A span $\mathbb{R}^n$ since if $v \in \mathbb{R}^n$ then $Au = v$ for $u = Bv \in \mathbb{R}^n$, so A is invertible. Therefore $B = A^{-1}AB = A^{-1}I_n = A^{-1}$ so $BA = A^{-1}A = I_n$. $\square$

Important note: this corollary only applies to *square matrices*.

3 Subspaces of $\mathbb{R}^n$

Let n be a positive integer. Remember that $0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^n$.

Definition. Let H be a subset of $\mathbb{R}^n$. The subset H is a **subspace** if these three conditions hold:

1. $0 \in H$. ↪ Every subspace is non empty.
2. $u + v \in H$ for all $u, v \in H$. ↪ subspace is closed under vector addition
3. $cv \in H$ for all $c \in \mathbb{R}$ and $v \in H$. ↪ subspace is closed under scalar multiplication

Common examples

$\mathbb{R}^n$ is a subspace of itself.

The set $\{0\}$ consisting of just the zero vector is a subspace of $\mathbb{R}^n$.

The empty set $\emptyset$ is *not* a subspace since it does not contain the zero vector.

A subset $H \subseteq \mathbb{R}^2$ is a subspace if and only if $H = \{0\}$ or $H = \mathbb{R}^2$ or $H = \mathbb{R}\text{-span}\{v\}$ for some $v \in \mathbb{R}^2$

The span of a set of vectors in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$.

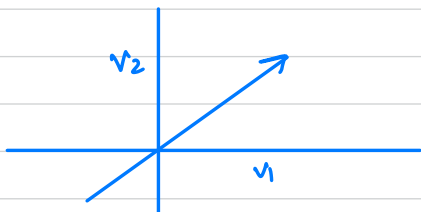
Conversely, any subspace of $\mathbb{R}^n$ is the span of a finite set of vectors, although this is not obvious.

↪ largest subspace
 "zero subspace" it has only one element
 ↪ every other subspace is infinite
 ↪ Smallest subspace

Example: The only subspaces of $\mathbb{R}^1 = \mathbb{R}$ are $\{0\}$ and $\mathbb{R}^1$.

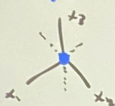
pf: If H is a subspace of $\mathbb{R}^1$ and $0 \neq x \in H$ then $cx \in H$ for all $c \in \mathbb{R}$ which means $y \in H$ for all $y \in \mathbb{R}^1$.

Example: Subspaces of $\mathbb{R}^2$ include $\{0\}$, $= \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$, $\mathbb{R}^2$ itself, and $\mathbb{R}\text{-span} \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} = H$

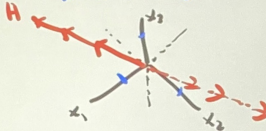


Ex Subspaces of $\mathbb{R}^3$

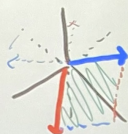
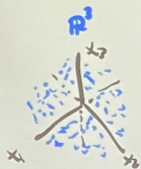
$\{0\}$



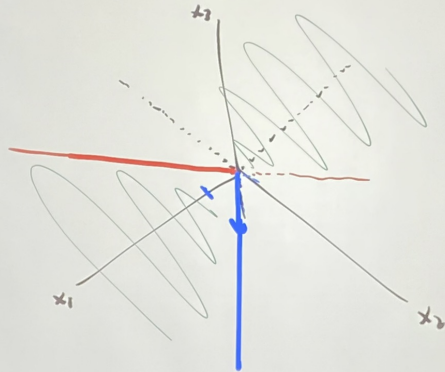
$H = \mathbb{R}\text{-span} \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \neq 0 \right\}$



$H = \mathbb{R}\text{-span} \{u, v\}$



What does $H = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3 \mid v_1 + v_2 + v_3 = 0 \right\}$ look like?



Example. The set

$$X = \left\{ v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 1 \right\}$$

is *not* a subspace since $0 \notin X$.

Example. The set

$$H = \left\{ v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0 \right\}$$

is a subspace since if $u, v \in H$ and $c \in \mathbb{R}$ then

$$(u_1 + v_1) + (u_2 + v_2) + (u_3 + v_3) = (u_1 + u_2 + u_3) + (v_1 + v_2 + v_3) = 0 + 0 = 0$$

and

$$cu_1 + cu_2 + cu_3 = c(u_1 + u_2 + u_3) = 0$$

so $u + v \in H$ and $cu \in H$.

Any matrix A gives rise to two subspaces, called the *column space* and *null space*.

Definition. The *column space* of an $m \times n$ matrix A is the subspace

col A $\subseteq \mathbb{R}^m$, $m = \#$ of rows of A .

$$\text{Col } A = \{ Ax : x \in \mathbb{R}^n \} \subseteq \mathbb{R}^m.$$

The set $\text{Col } A$ is the span of the columns of A .

Example. If $V = \mathbb{R}\text{-span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ then what are some matrices A with $\text{Col } A = V$?

Here are four examples:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 2 & 1 \\ 1 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}.$$

Many different matrices can have the same column space, and it may not be at all obvious whether a subspace V is equal to the column space of a given matrix A .

Remark. If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the linear function $T(x) = Ax$ then $\text{Col } A = \text{range}(T)$.

A vector $b \in \mathbb{R}^m$ belongs to $\text{Col } A$ if and only if $Ax = b$ has a solution.

Thus $\text{Col } A = \mathbb{R}^m$ if and only if $Ax = b$ has a solution for each $b \in \mathbb{R}^m$ ($\Leftrightarrow A$ has a pivot in every row).

Definition. The *null space* of an $m \times n$ matrix A is the subspace

$n = \#$ of columns

$$\text{Nul } A = \{ v \in \mathbb{R}^n : Av = 0 \} \subseteq \mathbb{R}^n$$

The set $\text{Nul } A$ is exactly the set of solutions to the matrix equation $Ax = 0$.

Proof that $\text{Nul } A$ is a subspace. If $u, v \in \text{Nul } A$ and $c \in \mathbb{R}$ then $A(u + v) = Au + Av = 0 + 0 = 0$ and $A(cv) = c(Av) = 0$, so $u + v \in \text{Nul } A$ and $cv \in \text{Nul } A$. Thus $\text{Nul } A$ is a subspace of $\mathbb{R}^n$. $\square$

Easy reason why a basis for $H \subseteq \mathbb{R}^n$
must have $\leq n$ elements:

* any set of more than n vectors in $\mathbb{R}^n$
is linearly dependent

If $H = \{0\} \subseteq \mathbb{R}^n$ but its basis has 0 elements

If $H = \text{span}\{e_1\} = \left\{ \begin{bmatrix} x \\ 0 \\ \vdots \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\} \subseteq \mathbb{R}^n$

has basis $\{e_1\}$ with just one element

Remark. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the linear function $T(x) = Ax$ then $\text{Nul } A = \{x \in \mathbb{R}^n : T(x) = 0\}$.

The **column space** is a subspace of $\mathbb{R}^m$ where m is the **number of rows of A** .

The **null space** is a subspace of $\mathbb{R}^n$ where n is the **number of columns of A** .

A subspace can be completely determined by a finite amount of data. This data will be called a **basis**.

Definition. Let H be a subspace of $\mathbb{R}^n$. A **basis** for H is a set of **vectors** $\{v_1, v_2, \dots, v_k\} \subseteq H$ that are **linearly independent** and have **span equal to H** .

The empty set $\emptyset = \{\}$ is considered to be a basis for the zero subspace $\{0\}$.

Example. The set $\{e_1, e_2, \dots, e_n\} \subseteq \mathbb{R}^n$ where $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, and so on, is a basis for $\mathbb{R}^n$.

We call this the **standard basis** of $\mathbb{R}^n$.

Theorem. Every subspace H of $\mathbb{R}^n$ has a basis of size at most n .

Proof. If $H = \{0\}$ then $\emptyset$ is a basis.

Assume $H \neq \{0\}$. Let $\mathcal{B}$ be a set of linearly independent vectors in H that is as large as possible. The size of $\mathcal{B}$ must be at most n since any $n+1$ vectors in $\mathbb{R}^n$ are linearly dependent.

Let $w_1, w_2, \dots, w_k$ be the elements of $\mathcal{B}$. Since $\mathcal{B}$ is as large as possible, if $v \in H$ is any vector then $w_1, w_2, \dots, w_k, v$ are linearly dependent so we can write

$$c_1 w_1 + c_2 w_2 + \dots + c_k w_k + cv = 0$$

for some numbers $c_1, c_2, \dots, c_k, c \in \mathbb{R}$ which are not all zero.

If $c = 0$ then this would imply that the vectors in $\mathcal{B}$ are linearly dependent. But the vectors in $\mathcal{B}$ are linearly independent, so we must have $c \neq 0$. Therefore

$$v = \frac{c_1}{c} w_1 + \frac{c_2}{c} w_2 + \dots + \frac{c_k}{c} w_k.$$

This means that v is in the span of the vectors in $\mathcal{B}$. Since $v \in H$ is an arbitrary vector, we conclude that the span of the vectors in $\mathcal{B}$ is all of H , so $\mathcal{B}$ is a basis for H . $\square$

Example. Let $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$.

How can we find a basis for $\text{Nul } A$? Well, finding a basis for $\text{Nul } A$ is more or less the same task as finding all solutions to the homogeneous equation $Ax = 0$. So let's first try to solve that equation.

If we row reduce the 3×6 matrix $[A \ 0]$, we get

$$[A \ 0] \sim \begin{bmatrix} \textcircled{1} & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & \textcircled{1} & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \text{RREF}([A \ 0]).$$

2 basic variables, 3 free variables.

This tells us that $Ax = 0$ if and only if $\begin{cases} x_1 - 2x_2 - x_4 + 3x_5 = 0 \\ x_3 + 2x_4 - 2x_5 = 0 \end{cases}$ or equivalently $\begin{cases} x_1 = 2x_2 + x_4 - 3x_5 \\ x_3 = -2x_4 + 2x_5 \end{cases}$

By substituting these formulas for the basic variables x_1 and x_3 , we deduce that $x \in \text{Nul } A$ if and only if

only use free variables $\leftarrow$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

The vectors

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

① These vectors do span Nul A
② Also linearly independent.

are a basis for $\text{Nul } A$: we just computed that these vectors span the null space, and they are linearly independent since each has a nonzero entry in a row (namely, either row 2, 4, or 5) where the others have zeros. (Why does this imply linear independence?)

This example is important: the procedure just described works to construct a basis of $\text{Nul } A$ for any matrix A . **The size of this basis will always be equal to the number of free variables in the linear system $Ax = 0$.** How to find a basis for $\text{Nul } A$ is something you should learn and remember.

Example. Let $B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

This matrix is in reduced echelon form. How to find a basis for $\text{Col } B$?

The columns of B automatically span $\text{Col } B$, but they might not be linearly independent.

The largest linearly independent subset of the columns of B will be a basis for $\text{Col } B$, however.

In our example, the pivot columns 1, 2 and 5 are linearly independent since each has a row with a 1 where the others have 0s. These columns span columns 3 and 4, so a basis for $\text{Col } B$ is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

This example was special since the matrix B was already in reduced echelon form. To find a basis of the column space of an arbitrary matrix, we rely on the following observation:

Proposition. Let A be any matrix. The pivot columns of A form a basis for $\text{Col } A$.

Proof. Let $v_1, v_2, \dots, v_n$ be the columns of $A = [v_1 \ v_2 \ \dots \ v_n]$.

Consider the matrices $A_k = [v_1 \ v_2 \ \dots \ v_k]$ for $k = 1, 2, \dots, n$.

Observe that $\text{RREF}(A_k)$ is equal to **the first k columns** of $\text{RREF}(A)$.

If k is not a pivot column of A , then the last column of A_k is not a pivot column.

This means that $A_{k-1}x = v_k$ is consistent so v_k is in the span of $v_1, v_2, \dots, v_{k-1}$.

Thus each non-pivot column of A is a linear combination of earlier columns. This means that each non-pivot column of A is a linear combination of earlier columns **that are pivot columns**: if i_1 is the first non-pivot column, then v_{i_1} is a linear combination of earlier columns, which are all pivots; if i_2 is the second non-pivot column, then v_{i_2} is a linear combination of earlier columns, and these are all either

pivots or v_{i_1} , but in any linear combination involving v_{i_1} we can replace v_{i_1} by a linear combination of pivot columns to get a linear combination involving only pivot columns; if i_3 is the third non-pivot column, then v_{i_3} is a linear combination of earlier columns, and these are all either pivots or v_{i_1} or v_{i_2} , and we can replace v_{i_1} and v_{i_2} by combinations of pivot columns as needed; and so on.

We conclude that **Col A is spanned by the pivot columns of A** . Why are they linearly independent?

If k is a pivot column of A , then the last column of A_k is a pivot column.

This means that $A_{k-1}x = v_k$ is inconsistent so v_k is not in the span of $v_1, v_2, \dots, v_{k-1}$.

Therefore v_k is also not in the span of the (smaller) set of earlier columns **that are pivot columns**.

Thus if $j_1 < j_2 < \dots < j_q$ are the pivot columns of A then we have a strictly increasing chain of subspaces

$$\mathbb{R}\text{-span}\{v_{j_1}\} \subsetneq \mathbb{R}\text{-span}\{v_{j_1}, v_{j_2}\} \subsetneq \mathbb{R}\text{-span}\{v_{j_1}, v_{j_2}, v_{j_3}\} \subsetneq \dots \subsetneq \mathbb{R}\text{-span}\{v_{j_1}, v_{j_2}, \dots, v_{j_q}\}.$$

The fact that this chain is strictly increasing means $v_{j_1}, v_{j_2}, \dots, v_{j_q}$ are **also linearly independent**. $\square$

Example. The matrix

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix} \xrightarrow{\text{REF}(A)} \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is row equivalent to the matrix B in the previous example. Columns 1, 2, and 5 of A have pivots, so

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -9 \\ 2 \\ 1 \\ -8 \end{bmatrix} \right\}$$

is a **basis for Col A** .

These are usually not the only columns of A that form a basis for Col A .

Next time: we will show that if H is a subspace of $\mathbb{R}^n$ then all of its bases have the same size. The common size of each basis is the *dimension* of H .

4 Vocabulary

Keywords from today's lecture:

1. Subspace of $\mathbb{R}^n$

A subset $H \subseteq \mathbb{R}^n$ such that $0 \in H$; if $u, v \in H$ then $u + v \in H$; and if $v \in H$, $c \in \mathbb{R}$ then $cv \in H$.

Example: Pick any vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$. Then $\mathbb{R}\text{-span}\{v_1, v_2, \dots, v_p\}$ is a subspace.

2. Column space of an $m \times n$ matrix A .

The subspace $\text{Col } A = \{Av : v \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$. The span of the columns of A .

Example: If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ then $\text{Col } A = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \in \mathbb{R}^3 : x, y \in \mathbb{R} \right\}$.

3. Null space of an $m \times n$ matrix A .

The subspace $\text{Nul } A = \{v \in \mathbb{R}^n : Av = 0\} \subseteq \mathbb{R}^n$.

Example: If $A = \begin{bmatrix} 1 & -2 & 0 \\ -1 & 2 & 0 \end{bmatrix}$ then $\text{Nul } A = \left\{ \begin{bmatrix} 2x \\ x \\ y \end{bmatrix} \in \mathbb{R}^3 : x, y \in \mathbb{R} \right\} = \mathbb{R}\text{-span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

4. Basis of a subspace $H \subseteq \mathbb{R}^n$

A set of linearly independent vectors in H whose span is H .

Example: The vectors $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$ are a basis for the subspace $\left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0 \right\}$.

The **standard basis** of $\mathbb{R}^n$ consists of the vectors $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$.